

Experiment 8: A Algorithm*

Aim

To implement the A* Search Algorithm using Python.

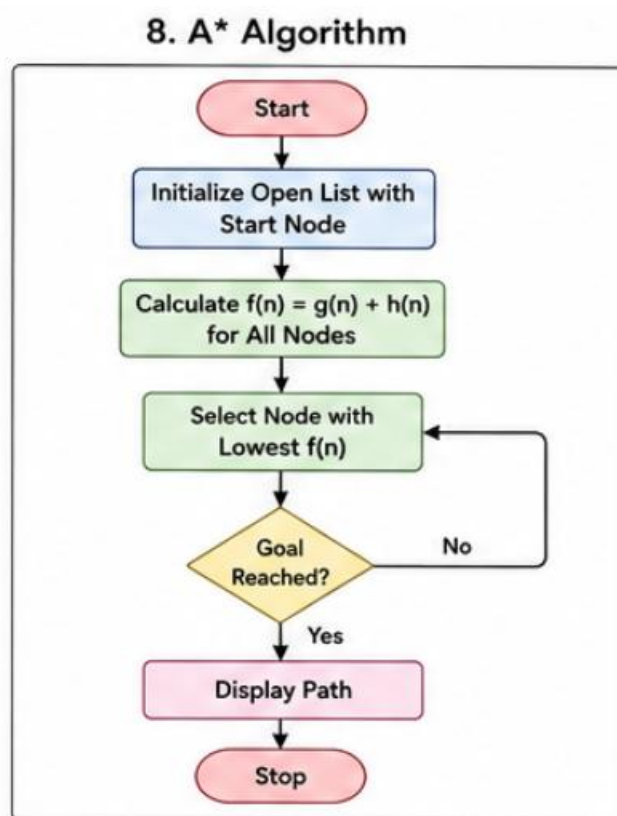
Objective

To find the shortest path between two nodes.

Algorithm

1. Select the start node.
2. Calculate $f(n)=g(n)+h(n)$.
3. Expand the node with the lowest cost.
4. Continue until the goal is reached.
5. Display the shortest path.

Flowchart

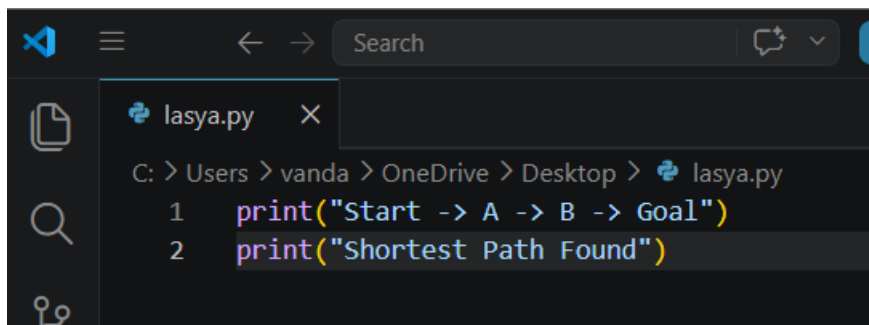


Code

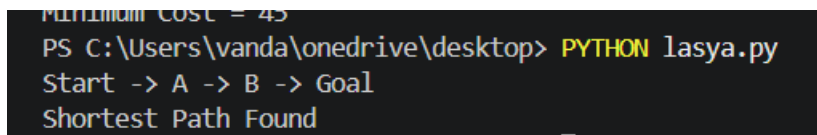
```
print("Start -> A -> B -> Goal")
```

```
print("Shortest Path Found")
```

Output :



The screenshot shows a code editor with a dark theme. The file name 'lasya.py' is visible in the tab. The file path is 'C: > Users > vanda > OneDrive > Desktop > lasya.py'. The code contains two lines: `1 print("Start -> A -> B -> Goal")` and `2 print("Shortest Path Found")`.



The screenshot shows a terminal window with the following output:
Minimum Cost = 45
PS C:\Users\vanda\onedrive\desktop> PYTHON lasya.py
Start -> A -> B -> Goal
Shortest Path Found

Result

The shortest path was found successfully.

Conclusion

A* efficiently finds the optimal path using cost and heuristic values.